

pipeline metric results

Benchmark: 30 Standardized Clinical Questions | **Corpus:** Pages 22–51 (46 Chunks) from Ackley & Ladwig's Nursing Diagnosis Handbook
Evaluation Suite: Information Retrieval (Precision@k, Recall@k, MRR, MAP, nDCG@5) | Context Quality | Grounded LLM Generation (Gemini 3.5 Flash-Lite)

P1: Baseline Plain BM25 + FAISS	P2: + Concepts SNOMED/MeSH Concepts	P3: + Intent Phase 8 Intent Rerank	P4: Contextual Embeddings (Fake Qs) Augmented: Context + Synthetic Qs	P5: Contextual Embeddings + HyDE HyDE + Augmented Multi-Vector
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1. Information Retrieval Metrics (5-Pipeline Comparison)

Metric	P1: Baseline	P2: + Concepts	P3: + Intent	P4: Contextual Embeddings (Fake Qs)	P5: Contextual Embeddings + HyDE	Δ (P4 vs Base)	Δ (P5 vs Base)
Precision@1	0.4000	0.3333	0.4000	0.4667	0.4667	+6.7%	+6.7%
Precision@3	0.2333	0.2111	0.2556	0.3444	0.3444	+11.1%	+11.1%
Precision@5	0.2000	0.1800	0.2133	0.2733	0.2667	+7.3%	+6.7%
Recall@1	0.1889	0.1667	0.2111	0.2611	0.2611	+7.2%	+7.2%
Recall@3	0.3222	0.2944	0.3667	0.5500	0.5500	+22.8%	+22.8%
Recall@5	0.4611	0.4222	0.4889	0.7222	0.7111	+26.1%	+25.0%
MRR	0.4639	0.4444	0.4722	0.5978	0.6000	+13.4%	+13.6%
MAP	0.3323	0.2951	0.3536	0.5008	0.5083	+16.9%	+17.6%
nDCG@5	0.4021	0.3662	0.4240	0.5853	0.5867	+18.3%	+18.5%

2. Context Quality & Prompt Sufficiency Metrics (Top-5 Injected Evidence)

Metric	P1: Baseline	P3: + Intent	P4: Contextual Embeddings (Fake Qs)	P5: Contextual Embeddings + HyDE	Δ (P4 vs Base)	Δ (P5 vs Base)
Context Recall	0.4611	0.4889	0.7222	0.7111	+26.1%	+25.0%
Context Precision	0.2000	0.2133	0.2733	0.2667	+7.3%	+6.7%
Context Sufficiency	0.6000	0.6333	0.8667	0.8333	+26.7%	+23.3%
Context Noise Ratio	0.8000	0.7867	0.7267	0.7333	-7.3%	-6.7%

3. LLM Generation Quality & Grounding (Gemini 3.5 Flash-Lite)

Metric	P1: Baseline	P3: + Intent	P4: Contextual Embeddings (Fake Qs)	P5: Contextual Embeddings + HyDE	Δ (P4 vs Base)	Δ (P5 vs Base)
ROUGE-1 F1	0.3128	0.3142	0.3373	0.3531	+2.4%	+4.0%
ROUGE-2 F1	0.1329	0.1310	0.1490	0.1579	+1.6%	+2.5%
ROUGE-L F1	0.2400	0.2446	0.2539	0.2669	+1.4%	+2.7%
BLEU-1	0.2274	0.2302	0.2373	0.2532	+1.0%	+2.6%
BLEU-2	0.1434	0.1398	0.1535	0.1624	+1.0%	+1.9%
BLEU-4	0.0499	0.0518	0.0600	0.0697	+1.0%	+2.0%
Citation Precision	1.0000	1.0000	1.0000	1.0000	0.0%	0.0%
Citation Coverage	0.5222	0.5722	0.7944	0.7500	+27.2%	+22.8%
Faithfulness	0.9778	0.9833	1.0000	1.0000	+2.2%	+2.2%
Relevance	0.7441	0.7545	0.6910	0.6999	-5.3%	-4.4%

4. Key Findings & Insights

- Contextual Embeddings (Fake Queries) Lead in Recall & Sufficiency:** Indexing synthetic clinical questions alongside chunk text and abstract context achieves peak Recall@5 (**0.7222** vs. 0.4611 Baseline, **+25.0%**) and Prompt Sufficiency (**0.8667** vs. 0.6000 Baseline, **+23.3%**).
- HyDE Blended Retrieval Maximizes Ranking Accuracy:** Blending hypothetical textbook passages with the clinical query yields the highest MRR (**0.6000** vs. 0.4639, **+13.6%**), MAP (**0.5083** vs. 0.3323, **+17.6%**), and nDCG@5 (**0.5867** vs. 0.4021, **+18.5%**).
- Phase 8 Intent Reranking:** Enhances baseline MRR to **0.4722** and Recall@3 to **0.3667** via intent-relevance multiplicative gating and topical entity anchoring.
- Zero Hallucinations Achieved:** Both Contextual Embeddings (P4) and HyDE (P5) achieve **100.0% Faithfulness** with 100% Citation Precision.